

Math 1232 Summer 2025

Midterm Exam

- You will have 90 minutes for this test.
- You are not allowed to consult books or notes during the test, but you may use a one-page, one-sided, handwritten cheat sheet you have made for yourself ahead of time.
- You may not use a calculator. You may leave answers unsimplified, except you should compute trigonometric functions as far as possible.
- The exam has 6 problems, one on each mastery topic we've covered. The exam has 6 pages total.
- Each part of each topic is worth ten points. The whole test is scored out of 100 points.
- Read the questions carefully and make sure to answer the actual question asked. Make sure to justify your answers—math is largely about clear communication and argument, so an unjustified answer is much like no answer at all. When in doubt, show more work and write complete sentences.
- If you need more paper to show work, I have extra at the front of the room.
- Good luck!

	a	b	c
M1:			
M2:			
S1:		S2:	
S3:		S4:	

Total /100

Name: **Solutions**

Problem 1 (M1).

(a) Compute $\int_1^2 \frac{e^{1/x}}{x^2} dx$.

Solution. We take $u = 1/x$ so $du = \frac{-1}{x^2} dx$. Then

$$\begin{aligned}\int_1^2 \frac{e^{1/x}}{x^2} dx &= \int_1^{1/2} -e^u du \\ &= -e^u \Big|_1^{1/2} = -e^{1/2} + e^1 = e - \sqrt{e}.\end{aligned}$$

(b) Compute $\frac{d}{dx}(x^3 - x)^{\arctan(x)}$.

Solution.

$$\begin{aligned}\ln |y| &= \arctan(x) \ln |x^3 - x| \\ \frac{y'}{y} &= \frac{\ln |x^3 - x|}{1 + x^2} + \frac{(3x^2 - 1) \arctan(x)}{x^3 - x} \\ y' &= y \left(\frac{\ln |x^3 - x|}{1 + x^2} + \frac{(3x^2 - 1) \arctan(x)}{x^3 - x} \right) \\ &= (x^3 - x)^{\arctan(x)} \left(\frac{\ln |x^3 - x|}{1 + x^2} + \frac{(3x^2 - 1) \arctan(x)}{x^3 - x} \right).\end{aligned}$$

(c) Compute $\frac{d}{dx} \pi^{\sqrt{x}}$.

Solution.

$$\frac{d}{dx} \pi^{\sqrt{x}} = \pi^{\sqrt{x}} \ln(\pi) \frac{1}{2\sqrt{x}}.$$

Problem 2 (M2). Compute the following integrals using any of the methods we have learned in class. Show enough work to justify your answers.

(a) $\int \frac{\sqrt{1-x^2}}{x^2} dx.$

Solution. We're going to set $x = \sin \theta$ so that $dx = \cos \theta d\theta$. Then

$$\begin{aligned}\int \frac{\sqrt{1-x^2}}{x^2} dx &= \int \frac{\sqrt{1-\sin^2(\theta)}}{\sin^2(\theta)} \cos(\theta) d\theta \\ &= \int \frac{\cos^2(\theta)}{\sin^2(\theta)} d\theta \\ &= \int \cot^2(\theta) d\theta \\ &= \int \csc^2(\theta) - 1 d\theta \\ &= -\cot(\theta) - \theta + C.\end{aligned}$$

But we know that $\sin(\theta) = x$, so $x = \arcsin(\theta)$ and $\cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)} = \frac{\sqrt{1-x^2}}{x}$. Then we get

$$\int \frac{\sqrt{1-x^2}}{x^2} dx = -\frac{\sqrt{1-x^2}}{x} - \arcsin(x) + C.$$

(b) $\int_{\pi/9}^{\pi/6} x \cos(3x) dx$

Solution.

$$\begin{aligned}\int_{\pi/9}^{\pi/6} x \cos(3x) dx &= \frac{1}{3} x \sin(3x) \Big|_{\pi/9}^{\pi/6} - \int_{\pi/9}^{\pi/6} \frac{1}{3} \sin(3x) dx \\ &= \frac{\pi}{18} \sin(\pi/2) - \frac{\pi}{27} \sin(\pi/3) + \frac{1}{9} \cos(3x) \Big|_{\pi/9}^{\pi/6} \\ &= \frac{\pi}{18} - \frac{\pi\sqrt{3}}{54} + \frac{1}{9} \cos(\pi/2) - \frac{1}{9} \cos(\pi/3) \\ &= \frac{\pi}{18} - \frac{\pi\sqrt{3}}{54} - \frac{1}{18}.\end{aligned}$$

(c) Compute $\int \frac{4x^2 - x + 10}{(x - 2)(x^2 + 4)} dx$.

Solution. We need to do a partial fractions decomposition. We have

$$\frac{4x^2 - x + 10}{(x - 2)(x^2 + 4)} = \frac{A}{x - 2} + \frac{Bx + C}{x^2 + 4}$$

$$4x^2 - x + 10 = A(x^2 + 4) + (Bx + C)(x - 2).$$

$$x = 2 : \quad 24 = A \cdot 8$$

$$A = 3$$

$$x = 0 : \quad 10 = 4A + (C)(-2) = 12 - 2C$$

$$C = 6 - 5 = 1$$

$$x = 1 : \quad 13 = 3(1 + 4) + (B + 1)(1 - 2) = 15 - B - 1$$

$$B = 1.$$

Thus we compute

$$\begin{aligned} \int \frac{4x^2 - x + 10}{(x - 2)(x^2 + 4)} dx &= \int \frac{3}{x - 2} + \frac{x + 1}{x^2 + 4} dx \\ &= \int \frac{3}{x - 2} + \frac{x}{x^2 + 4} + \frac{1}{x^2 + 4} dx \\ &= 3 \ln |x - 2| + \frac{1}{2} \ln |x^2 + 4| + \frac{1}{4} \int \frac{1}{(x/2)^2 + 1} dx \\ &= 3 \ln |x - 2| + \frac{1}{2} \ln |x^2 + 4| + \frac{1}{2} \arctan(x/2) + C. \end{aligned}$$

Name: [Solutions](#)

Problem 3 (S1). Let $h(x) = e^{x^3+x}$. Compute $(h^{-1})'(e^2)$.

Solution. By the Inverse Function Theorem, we know that

$$(h^{-1})'(e^2) = \frac{1}{h'(h^{-1}(e^2))}.$$

Guess and check shows that $h(1) = e^2$ so $h^{-1}(e^2) = 1$. And we know that

$$h'(x) = e^{x^3+x}(3x^2 + 1)$$

and thus

$$h'(1) = e^2(3 + 1) = 4e^2.$$

Thus

$$(h^{-1})'(5) = \frac{1}{4e^2}.$$

Problem 4 (S2). Compute $\lim_{x \rightarrow 0} \frac{7^x - 3^{2x}}{x}$.

Solution. The top and bottom both approach zero, so we can use L'Hôpital's rule.

$$\lim_{x \rightarrow 0} \frac{7^x - 3^{2x}}{x} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0} \frac{\ln(7)7^x - 2\ln(3)3^{2x}}{1} = \ln(7) - 2\ln(3).$$

Problem 5 (S3). Compute $\int_1^\infty \frac{dx}{\sqrt{x} + x\sqrt{x}}$.

Solution. We'll take $u = \sqrt{x}$ so $du = \frac{dx}{2\sqrt{x}}$ and $\frac{dx}{\sqrt{x} + x\sqrt{x}} = \frac{2du}{1+u^2}$. Then

$$\begin{aligned}\int_1^\infty \frac{dx}{\sqrt{x} + x\sqrt{x}} &= \lim_{t \rightarrow \infty} \int_1^t \frac{dx}{\sqrt{x} + x\sqrt{x}} \\ &= \lim_{t \rightarrow \infty} \int_1^{\sqrt{t}} \frac{2du}{1+u^2} \\ &= \lim_{t \rightarrow \infty} 2 \arctan(u) \Big|_1^{\sqrt{t}} = \lim_{t \rightarrow \infty} 2 \arctan(\sqrt{t}) - 2 \arctan(1) \\ &= 2(\pi/2) - 2(\pi/4) = \pi/2.\end{aligned}$$

Problem 6 (S4). Set up (but don't compute!) an integral for the area of a surface obtained by taking the curve $y = \ln(x^3 + 1)$ from $x = 0$ to $x = 10$ and rotating around the x -axis.

Solution. We have $y' = \frac{3x^2}{x^3+1}$ so we get

$$\begin{aligned}SA &= \int_0^{10} 2\pi y \sqrt{1 + y'^2} \, dx \\ &= \int_0^{10} 2\pi \ln(x^3 + 1) \sqrt{1 + \frac{9x^4}{(x^3 + 1)^2}} \, dx.\end{aligned}$$